

A Norm-One Extension Lifting Principle for c_0 in Pointwise Norm-Attaining Lipschitz Functions

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Status

This packet gives a likely valid partial result toward Question 4.1 of Choi–Jung–Lee–Roldan [1]. The source asks whether c_0 is isometrically contained in $\text{PNA}(M)$ for every infinite pointed metric space M . The theorem below proves an exact permanence principle: the answer is positive whenever a positive subspace $K \subseteq M$ admits a norm-one linear Lipschitz extension operator into M . In particular, it is positive when such a K is a pointed 1-Lipschitz retract of M .

Source Question

Question 4.1 in the source asks:

Is it true for every infinite pointed metric space M that c_0 is isometrically contained in $\text{PNA}(M)$?

4. EMBEDDING OF c_0 IN POINTWISE NORM-ATTAINING LIPSCHITZ FUNCTIONS

Let us start by recalling the facts (S6)–(S8), Theorem 2.9, and Theorem 3.9.

- For any infinite pointed metric space M , we have that c_0 is always isomorphically contained in $\text{SNA}(M)$. Moreover, if M is not uniformly discrete, then c_0 is isometrically contained in $\text{SNA}(M)$.
- However, there exist some metric spaces M such that c_0 is not isometrically contained in $\text{SNA}(M)$.
- There are metric spaces M such that ℓ_∞ is not isometrically contained in $\text{PNA}(M)$.

These motivate us to pose the following natural question.

Question 4.1. Is it true for every infinite pointed metric space M that c_0 is isometrically contained in $\text{PNA}(M)$?

In this section, we provide four distinct results, namely Proposition 4.2, Theorems 4.3, 4.6, and 4.7, all of which concern the isometric embedding of c_0 into $\text{PNA}(M)$ in the setting of a uniformly discrete metric space M . Consequently, we observe that for all the explicitly known metric spaces M such that c_0 is not isometrically embedded in $\text{SNA}(M)$, it is actually possible to embed c_0 isometrically in $\text{PNA}(M)$. Finally, we prove the main result, Theorem 4.11, which gives a partial answer to Question 4.1 by showing that for any infinite pointed metric space M there is a metric space $M_0 \subseteq M$ for which $\text{SNA}(M_0)$ contains an isometric copy of c_0 .

We start with the following analogue of Proposition 3.1 for pointwise norm-attaining Lipschitz functions. The proof of the result is an easy adaptation of Proposition 3.1 and [15, Lemma 3.3].

Proposition 4.2. *Let M be an infinite uniformly discrete pointed metric space, and let $\{p_n\}_{n=1}^\infty$ be a sequence of distinct points in M . The following claims are equivalent:*

- (a) $\text{PNA}(M)$ contains c_0 isometrically with a sequence of functions $\{f_n\}_{n=1}^\infty$ that are isometrically a canonical basis of c_0 inside $\text{PNA}(M)$ and given by

$$f_n(p_n) = a_n, \quad f_n(p) = 0 \quad \text{if } p \in M \setminus \{p_n\}$$

for some numbers $\{a_n\}_{n=1}^\infty \subseteq \mathbb{R}$.

- (b) For each $m \neq n \in \mathbb{N}$, the following holds

$$d(p_n, p_m) \geq R(p_n) + R(p_m).$$

Again, there are simple uniformly discrete spaces that do not satisfy the condition in Proposition 4.2, such as the space M where $d(x, y) = 1$ for all $x \neq y \in M$. However, this metric space M satisfies the assumption in Theorem 3.4; hence $\text{SNA}(M)$ contains an isometric copy of c_0 .

The source proves that every infinite M contains a metric subspace $M_0 \subseteq M$ such that c_0 is isometrically contained in $\text{PNA}(M_0)$. The remaining obstruction is the subspace-to-ambient lifting step: scalar norm-preserving Lipschitz extensions are available one function at a time, but they are not automatically linear on an entire c_0 -copy.

Definitions

Let M be a pointed metric space. The Banach space $\text{Lip}_0(M)$ consists of all scalar Lipschitz functions $f : M \rightarrow \mathbb{R}$ with $f(0) = 0$, equipped with the Lipschitz norm

$$\|f\|_L = \sup_{p \neq q} \frac{|f(p) - f(q)|}{d(p, q)}.$$

Following the source, $f \in \text{Lip}_0(M)$ is pointwise norm-attaining if there is some $p \in M$ such that

$$\sup_{q \in M \setminus \{p\}} \frac{|f(p) - f(q)|}{d(p, q)} = \|f\|_L.$$

The set of such functions is denoted by $\text{PNA}(M)$.

Proof Intuition

The source's subspace theorem almost solves the problem, except that Lipschitz extension from M_0 to M is usually nonlinear. A nonlinear McShane extension preserves each individual norm, but it does not preserve the linear c_0 -structure.

The obstruction disappears exactly when the relevant subspace has a simultaneous linear extension operator of norm one. If $J : c_0 \rightarrow \text{Lip}_0(K)$ is an isometric copy consisting of pointwise norm-attaining functions and $E : \text{Lip}_0(K) \rightarrow \text{Lip}_0(M)$ is a norm-one extension operator, then EJ is still linear. It is contractive because E has norm one, and it is isometric because restricting $EJ(a)$ back to K recovers $J(a)$. Finally, the same point in K that witnesses pointwise norm-attainment for $J(a)$ also witnesses it for $EJ(a)$, since the supremum over M contains the supremum over K .

Main Theorem

Theorem 1 (Norm-one extension lifting). *Let $K \subseteq M$ be pointed metric spaces with the same distinguished point. Suppose there is a linear operator*

$$E : \text{Lip}_0(K) \longrightarrow \text{Lip}_0(M)$$

such that $\|E\| = 1$ and $(Ef)|_K = f$ for every $f \in \text{Lip}_0(K)$. If $\text{PNA}(K)$ contains an isometric copy of c_0 , then $\text{PNA}(M)$ contains an isometric copy of c_0 .

Proof. Let $J : c_0 \rightarrow \text{Lip}_0(K)$ be a linear isometry such that

$$J(c_0) \subseteq \text{PNA}(K) \cup \{0\}.$$

Define $T = EJ : c_0 \rightarrow \text{Lip}_0(M)$. Since E is linear, T is linear. For every $a \in c_0$,

$$\|Ta\|_{\text{Lip}_0(M)} \leq \|E\| \|Ja\|_{\text{Lip}_0(K)} = \|a\|_\infty.$$

On the other hand, restriction to K gives

$$\|Ta\|_{\text{Lip}_0(M)} \geq \|(Ta)|_K\|_{\text{Lip}_0(K)} = \|Ja\|_{\text{Lip}_0(K)} = \|a\|_\infty.$$

Thus T is a linear isometry.

It remains to check pointwise norm-attainment. Let $a \in c_0 \setminus \{0\}$. Since $Ja \in \text{PNA}(K)$, there exists $p \in K$ such that

$$\sup_{q \in K \setminus \{p\}} \frac{|Ja(p) - Ja(q)|}{d(p, q)} = \|Ja\|_{\text{Lip}_0(K)} = \|a\|_\infty.$$

Because Ta extends Ja , the same slopes are present in M :

$$\sup_{q \in M \setminus \{p\}} \frac{|Ta(p) - Ta(q)|}{d(p, q)} \geq \sup_{q \in K \setminus \{p\}} \frac{|Ja(p) - Ja(q)|}{d(p, q)} = \|a\|_\infty.$$

The opposite inequality follows from $\|Ta\|_{\text{Lip}_0(M)} = \|a\|_\infty$. Hence $Ta \in \text{PNA}(M)$. Therefore $T(c_0) \subseteq \text{PNA}(M) \cup \{0\}$, as claimed. $\square$

Corollary 1 (1-Lipschitz retract case). *Let $K \subseteq M$ be pointed metric spaces. Suppose there is a 1-Lipschitz retraction $r : M \rightarrow K$ with $r|_K = \text{Id}_K$. If $\text{PNA}(K)$ contains an isometric copy of c_0 , then $\text{PNA}(M)$ contains an isometric copy of c_0 .*

Proof. Define $E : \text{Lip}_0(K) \rightarrow \text{Lip}_0(M)$ by $Ef = f \circ r$. This operator is linear, extends each f , and satisfies $\|E\| \leq 1$ because r is 1-Lipschitz. Since E is an extension operator, $\|E\| = 1$ unless $\text{Lip}_0(K) = \{0\}$, and the trivial case is irrelevant for containing c_0 . Theorem 1 applies. $\square$

Consequences for the Source Problem

Combining Corollary 1 with the source's positive classes gives ambient-space positive cases of Question 4.1. For example, if M contains an infinite pointed 1-Lipschitz retract K for which the source proves $c_0 \subseteq \text{PNA}(K)$ isometrically, then $c_0 \subseteq \text{PNA}(M)$ isometrically. This includes retracts K that are not uniformly discrete and the uniformly discrete classes covered by the source's Section 4 criteria.

Limitations

This packet does not prove that every source subspace $M_0 \subseteq M$ constructed in Theorem 4.10 admits such an extension operator. The theorem therefore leaves the full Question 4.1 open. Its role is to isolate a sharp and checkable extension hypothesis under which the subspace result upgrades to the ambient space.

Novelty Check

The local run indexes were searched for arXiv:2312.00393, pointwise norm-attainment, c_0 , 1-Lipschitz retracts, norm-one linear extension operators, and extension-lifting language. No duplicate packet or attempt was found. A bounded web search on June 29, 2026 for the same phrases found the source paper but no later explicit answer or matching extension-lifting statement.

Human Review Recommendation

Review this as a partial permanence result, not as a full solution of Question 4.1. The proof is short; the main review question is whether the norm-one linear extension hypothesis is useful enough to count as a substantive partial result toward the source's subspace-to-ambient obstruction.

References

- [1] G. Choi, M. Jung, H. J. Lee, and O. Roldan, *Embeddings of infinite-dimensional spaces in the sets of norm-attaining Lipschitz functions*, arXiv:2312.00393, 2023.